

M1-INH-E1b — submission bundle and independent verification invitation

Status (2026-10-09): Candidate **class-relative mathematical insufficiency theorem**, supported by an exact symbolic certificate; **external independent proof review pending**. This document is an external invitation, **not** acceptance of the theorem or a change to any canonical evidence/eligibility gate. The governing authority remains the [STAR Research Charter v0.2 PDF](#) and the [frozen M1-INH-E1 protocol](#).

Precisely claimed theorem (M1-INH-E1b)

Within the **registered nondegenerate**, $\beta_{,z} \neq 0$ Szekeres–Szafron irrotational dust + Λ class, the **complete frozen elliptic-evolution record**

$$\mathfrak{E}[S]|_J = (M, K, f, \Lambda, \epsilon, v_0, \sigma; g_2, g_3, \Delta_\wp)|_J$$

does not determine the full dust inhomogeneity up to diffeomorphism. The two explicit regular positive-density Einstein–dust solutions below have **identical $\mathfrak{E}$ component by component**, with the same inverse-Weierstrass representatives and time normalization, but are not related by a diffeomorphism of Einstein–dust solutions: their *curvature-defined* scalar $\mathcal{J}$ vanishes identically for the first and is positive on an open subset of the second. This is an **existence/insufficiency** claim for this registered class and record, **not** a universal no-go theorem.

Exact domain, metrics, signs and elliptic branch

Coordinates (t, z, x, y) , signature $(+ - - -)$, future dust flow $u = \partial_t$, convention

$$G_{ab} - \Lambda g_{ab} = \kappa \tilde{\rho} u_a u_b, \quad \Lambda = 3, \quad \mathcal{U} = \{1 < t < 2, -\frac{1}{4} < z < \frac{1}{4}, x^2 + y^2 < \frac{1}{4}\}, \quad J = (-\frac{1}{4}, \frac{1}{4}).$$

Set $M = 1 + z$, $K = 0$, $f = z$, $P = \Phi = [2(1 + z)]^{1/3} \sinh^{2/3}(\frac{3}{2}(t + z))$, $H = \coth(\frac{3}{2}(t + z))$, $B = [3(1 + z)]^{-1}$, $F = H^2 - 1$ and $q = e^{2z}(x^2 + y^2)$. For $k = 0, 1$, define

$$g^{(k)} = \text{diag}(1, -a_k^2, -c_k^2, -c_k^2), \quad a_k = P(H + B + d_k), \quad c_k = \frac{2Pw_k}{1 + w_k^2(x^2 + y^2)},$$

with $(w_0, d_0) = (1, 0)$ and $(w_1, d_1) = (e^z, (1 - q)/(1 + q))$. The elliptic-blind sectors correspond to $A = C = \frac{1}{2}$ and $(A, C) = (e^z/2, e^{-z}/2)$, respectively, with $B_1 = B_2 = 0$ and $h = 1$; each satisfies the registered spatial constraint. On $\mathcal{U}$, $P, B, F, H > 0$, $q < 1$, $a_k, c_k > 0$, and the curvature-derived quantities are

$$R_k := \text{tr}(G^{(k)a}_b - 3\delta_b^a) = \kappa \tilde{\rho}_k, \quad R_0 = \frac{3FB}{H + B} > 0, \quad R_1 = \frac{3F(B + d_1)}{H + B + d_1} > 0.$$

The **identical** frozen Weierstrass data are $g_2 = 0$, $g_3 = -M^2/4$, $\Delta_\wp = -27M^4/16 \neq 0$, $\epsilon = 0$, and the **same expanding real branch** $\dot{P} > 0$, $\wp'(u) < 0$, with positive-real inverse and period indices $(0, 0)$:

$$\xi = \frac{M}{2P}, \quad u(t, z) = \int_\xi^\infty \frac{d\eta}{\sqrt{4\eta^3 + M^2/4}}, \quad v_0(z) = \int_0^\infty \frac{d\eta}{\sqrt{4\eta^3 + M^2/4}}, \quad 0 < u < v_0, \quad t + z = \int_0^u P(u', z) du'.$$

These choices fix σ (common connected interval, expansion sign and inverse/lattice representatives); they are **not** inferred solely from equality of $(g_2, g_3, \Delta_\wp)$.

Intrinsic distinguishing certificate

For $R = \kappa \tilde{\rho} > 0$, the rank-one tensor $G^a_b - 3\delta_b^a$ uniquely determines the timelike dust eigendirection up to time orientation. Its nonzero shear σ^a_b determines the repeated-eigenvalue two-plane via $h^a_b = \delta_b^a - u^a u_b$, $I_j = \text{tr}(\sigma^j)$, and $P_\perp^a_b = \frac{2}{3}h^a_b - \frac{I_2}{3I_3}\sigma^a_b$. Hence $\mathcal{J} = -P_\perp^{ab}\nabla_a R \nabla_b R$ is a scalar invariant of the Einstein–dust solution. The claimed exact identities are

$$\mathcal{J}[g^{(0)}] \equiv 0, \quad \mathcal{J}[g^{(1)}] = \frac{36H^2F^2q}{P^2(H + B + d_1)^4(1 + q)^2} > 0 \quad \text{on } \mathcal{U} \cap \{x^2 + y^2 > 0\}.$$

Thus no open set in the second solution can be diffeomorphic (as an Einstein–dust solution) to an open set in the first on which $\mathcal{J}$ is identically zero.

Canonical verifier identity and replication

Canonical source: [m1_inh_e1b_metric_first_audit_v4.py](#) (Python 3; SymPy 1.14.0). **Expected exact file SHA-256:**

10ccfa264f475fd7bd6d490c80c8862a174730567ee51c0d759b87939b93e6bb.

Run in a *disposable checkout* (the verifier writes a JSON certificate beside itself): `sha256sum preregistrations/M1-INH-E1/evidence/metric-first/m1_inh_e1b_metric_first_audit_v4.py` then `python preregistrations/M1-INH-E1/evidence/metric-first/m1_inh_e1b_metric_first_audit_v4.py` after installing `sympy==1.14.0`. Compare the new output against the [preserved proof record](#) and [certificate](#); do **not** call a rerun of the same code independent implementation.

Explicit non-claims: This is **not** the general **M1-INH-E1a** Einstein-to-Weierstrass derivation or its independent acceptance; **not** a **M1-INH-E1c** test of sufficiency for any selected observable; **not** a terminal **P0-T001** failure or exhaustion of the five-parent P0 search frame; and **not** physical ACSC support, proof of BSD, a cosmological correspondence, or authorization to execute an experiment.

Invitation: Independent researchers are invited to **re-implement from the two metric tensors**, in another symbolic tensor system or an independently authored differential-geometric derivation, and check: the complete Einstein tensor and sign convention; positive-density regularity on $\mathcal{U}$; the full *frozen* branch record; dust/shear spectral uniqueness; and the $\mathcal{J}$ identities, including possible counterexamples. Please publish source/derivation, environment, exact source hashes, failed as well as successful checks, and an explicit scoped verdict. Criticism and falsification are equally welcome. **Any change to the repository's independent-review/registry gates requires its existing separate approval process; distribution in a tagged release does not satisfy or bypass that process.**